

Solutions to Problem Solving Class 3: Ethernet

Computer Communications – DE
Department of Communication Systems, LTH
ht 2006

1. Smallest size: 64 bytes (18 bytes header, 46 bytes data)
Largest size: 1518 bytes (18 bytes header, 1500 bytes data)
2. Ratio of useful data for the smallest packet: $\frac{46}{64} = 0.72$, and for the largest:
 $\frac{1500}{1518} = 0.99$.
3. Because the sending station must be able to sense a potential collision before the entire frame is sent. Otherwise, in the case of collision, the frame will be discarded in the false belief that the frame has been successfully received by the destination.
4. We assume a propagation speed of $\frac{2}{3} \cdot c$, and use $s = v \cdot t$. The bit length is

$$s = 1 \cdot 10^{-7} \cdot \frac{2}{3} \cdot 3 \cdot 10^8 = 20m.$$

5. Again make use of $s = v \cdot t$;

$$t = \frac{500}{\frac{2}{3} \cdot 3 \cdot 10^8} = 2.5\mu s.$$

6. Worst case means that data has to traverse the entire cable twice (time to collision + time to detection) before the collision is sensed. Thus the requested time is $2 \cdot 2.5\mu s = 5\mu s$.
7. The minimum size is $\frac{2 \cdot 500}{8 \cdot 1 \cdot 10^{-7} \cdot \frac{2}{3} \cdot 3 \cdot 10^8} = 6.25$ bytes.
8. Smallest frame size = 64 bytes = $64 \cdot 8$ bitar. Hence $t = \frac{64 \cdot 8}{10 \cdot 10^6} = 51.2\mu s$.
9. Denote by x the length of the cable. The two LANs will have an equal time to detection (of collision), $t_d = \frac{x}{0.6 \cdot c}$, dependent only on the propagation speed. A packet of length y will however be transmitted in $t_{10} = \frac{y}{10 \cdot 10^6}$ seconds in the 10Mbps LAN, and only in $t_{100} = \frac{y}{100 \cdot 10^6}$ seconds in the 100 Mbps LAN. The proportion of lost time over effective time will thus be 10 times larger for the 100 Mbps LAN.
10. 10 times shorter.

11. CSMA/CD performs best during low loads. The following scenario is probable during a too high load: A station sends a packet and senses that it collides. It backs off a certain time. As we have a high load, several other stations experience the same thing. They all back off. When their individual backoff times have expired, they attempt to use the medium again. There is now a large probability of new collisions, as several stations were previously forced to back off. Practically none gets to send. In the meantime the content of the outbuffer of each station grows, imposing an even higher load.
12. The four left-hand side digits (including the colon) and the dashes are *not* parts of the frame. The first 6 bytes is the destination address, 08 00 20 7c 94 1c, the following 6 bytes is the source address, 00 00 39 51 90 37, whereas the consecutive 2 bytes is the PDU Type/Length Field. Everything is written in hexadecimal code so the 08 00 of the PDU Type/Length Field means $8 \cdot 16^2 = 2048$ in decimal notation. $2048 > 1536$ so this field indicates a particular type (not length) of the PDU data.